

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I R. Akshaya (192524118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Akshaya

25+25+19+19+9

(97)

AP

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Question 1

Case Study - Vehicle Fault Diagnosis System

An automobile service center is designing a rule-based Logical Agent for preliminary vehicle fault diagnosis. The system's knowledge base contains:

- i. If there is EngineWarningLight AND UnusualNoise $\rightarrow$ Possible EngineFailureRisk
- ii. If there is EngineWarningLight AND LowOilPressure $\rightarrow$ Possible OilLeak
- iii. If there is UnusualNoise AND OverheatingSign $\rightarrow$ Possible CoolantSystemFailure
- iv. Facts about Vehicle X: EngineWarningLight = True, UnusualNoise = True, OverheatingSign = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible faults for Vehicle X. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Vehicle X has CoolantSystemFailure.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Hospital Emergency Ward Routing Agent

A hospital uses a FOL-based knowledge base to manage robot routing priority in emergency corridors.

- I. for-all x: Patient(x) AND CriticalCondition(x) $\rightarrow$ PriorityCorridor(x) [Rule 1]
- II. for-all x: PriorityCorridor(x) $\rightarrow$ CorridorCleared(x) AND NOT NormalQueue(x) [Rule 2]
- III. for-all x for-all y: Patient(x) AND WaitingBehind(x,y) AND CorridorCleared(y) $\rightarrow$ Hold(x) [Rule 3]
- IV. Facts: Patient(PatientA), CriticalCondition(PatientA), Patient(PatientB), WaitingBehind(PatientB, PatientA)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.
- (b) Using the Resolution method, prove that 'Hold(PatientB)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous critical patients. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Weather-Based Flight Scheduling Expert System

An airline scheduling logical agent operates with the following propositional knowledge base (KB).

- 1) P1: StormWarning $\rightarrow$ FlightDelayNeeded
- 2) P2: FlightDelayNeeded AND InternationalRoute $\rightarrow$ NotifyPassengers
- 3) P3: NOT NotifyPassengers AND InternationalRoute $\rightarrow$ ComplianceRisk
- 4) Facts: StormWarning = True, InternationalRoute = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move NOT inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'NotifyPassengers'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'StormWarning' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'ComplianceRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Cave Exploration Knowledge Agent

An AI agent exploring an underground cave grid perceives: Draft at [1,3], Glow at [1,1], Sparkle at [2,3]. The agent uses a propositional KB with these rules:

- 1) $\text{Draft}[1,3] \rightarrow \text{Pit is adjacent to cell } [1,3]$
- 2) $\text{Glow}[1,1] \rightarrow \text{LavaVent is adjacent to cell } [1,1]$
- 3) $\text{Sparkle}[x,y] \rightarrow \text{Treasure is at } [x,y]$
- 4) $\text{NOT Pit}[x,y] \text{ AND NOT LavaVent}[x,y] \rightarrow \text{Safe}[x,y]$

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Pit and LavaVent. Show each reasoning step precisely.
- (c) Evaluate whether the path $[1,1] \rightarrow [1,3] \rightarrow [2,3]$ is safe for the agent to collect the Treasure. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.



VEHICLE DIAGNOSIS SYSTEM.

(a) Propositional Representation:

Define:

E : Engine Warning light is true

U : Unusual Noise is true

O : Low Oil Pressure is true

H : Overheating sign is true.

F : Possible Engine Failure Risk.

Q : Possible Oil leak.

C : Possible Coolant System Failure.

Rules:

1. $E \wedge U \rightarrow F$

2. $E \wedge O \rightarrow Q$

3. $U \wedge H \rightarrow C$.

Facts for Vehicle $\rightarrow X$:

$E = \text{True}$, $U = \text{True}$, $H = \text{True}$ or simply:

E, U, H .

(b) Forward Chaining:

Initial facts : E, U, H .

Step 1: Apply Rule 1

$$E \wedge U \rightarrow F$$

Since E and U are true:

F

Therefore:

Possible Engine Failure Risk is derived.

Step 2: Apply Rule 2

$$E \wedge O \rightarrow Q$$

We have E , but there is no fact O (Low Oil Pressure).

Therefore: Rule 2 cannot fire.

Step 3: Apply Rule 3

$$U \wedge H \rightarrow C$$

Since both U and H are true:

C

Therefore:

Possible Coolant System Failure is derived.

$\therefore$ Possible Oil leak cannot be derived as not given.

(c) Backward chaining for Coolant System Failure:

Goal:

$C = \text{Possible Coolant System Failure}$

Find a new rule whose conclusion is C :

$$U \wedge H \rightarrow C$$

So the goal is reduced to two subgoals:

U and H

Both are facts: $U = \text{True}$, $H = \text{True}$.

$$\therefore U \wedge H \Rightarrow C$$

Hence: Possible Coolant System Failure

Goal reduction tree:

Possible Coolant System Failure

↓ Rule 3

$U \text{ AND } H$

U
↓
Fact

H
↓
Fact

So the goal is successfully proved.

2. HOSPITAL EMERGENCY WARD ROUTING AGENT:

(a) Unification for Rule 1:

Rule 1: Patient (x) $\wedge$ Critical Condition (x) $\rightarrow$ Priority

Facts:

Patient (Patient A)

Critical Condition (Patient A)

Unification

Match:

Patient (x)

with:

Patient (Patient A)

$$\therefore \theta_1 = \{x / \text{Patient}\}$$

Now check: Critical Condition (x)

After substitution: Critical Condition (Patient A)

This matches the given fact

$$\therefore \theta = \{x / \text{Patient A}\}$$

Applying substitution gives Patient (Patient A)
 $\wedge$ Critical Condition (Patient A) $\therefore$ It is derived.

6) Resolution proof of Hld (Patient B).

Relevant Rules:

Rule 1:

Patient (x) $\wedge$ Critical Condition (x) $\rightarrow$ Priority (x)

CNF:

$\neg$ Patient (x) $\vee \neg$ Critical Condition (x) $\vee$ Priority (x)

Rule 2

Priority Corridor (x) $\leftrightarrow$ Corridor Cleared (x) $\wedge \neg$
Normal Queue (x)

This gives:

$\neg$ Priority Corridor (x) $\vee$ Corridor Cleared (x)

and

$\neg$ Priority Corridor (x) $\vee \neg$ Normal Queue (x)

Rule 3

Patient (x) $\wedge$ Waiting Behind (x, y) $\wedge$ Corridor
cleared (y) $\rightarrow$ Hld (x)

CNF:

$\neg$ Patient (x) $\vee \neg$ Waiting Behind (x, y) $\vee$ Corridor
cleared (y) $\vee$ Hld (x)

CAVE EXPLORATION KNOWLEDGE AGENT:

(a) Belief State :

Initial facts:

Draft [1, 3], Glow [1, 1], Sparkle [2, 3]

Using Modus Ponens :

Draft [1, 3] $\rightarrow$ Pit Adjacent [1, 3]

Glow [1, 1] $\rightarrow$ LavaVent Adjacent [1, 1]

Sparkle [2, 3] $\rightarrow$ Treasure [2, 3].

(b) Possible Locations:

Draft [1, 3] $\Rightarrow$ Pit adjacent to [1, 3]

Pit = { [1, 2], [2, 3] }

LavaVent: Glow [1, 1] $\Rightarrow$ LavaVent adjacent to [1, 1]

(c) Path Safety:

[1, 1] $\rightarrow$ [1, 3] $\rightarrow$ [2, 3]

Since [2, 3] is possible pit location, safety cannot be guaranteed.

It is needed to confirm that [2, 3] has no Pit or LavaVent.

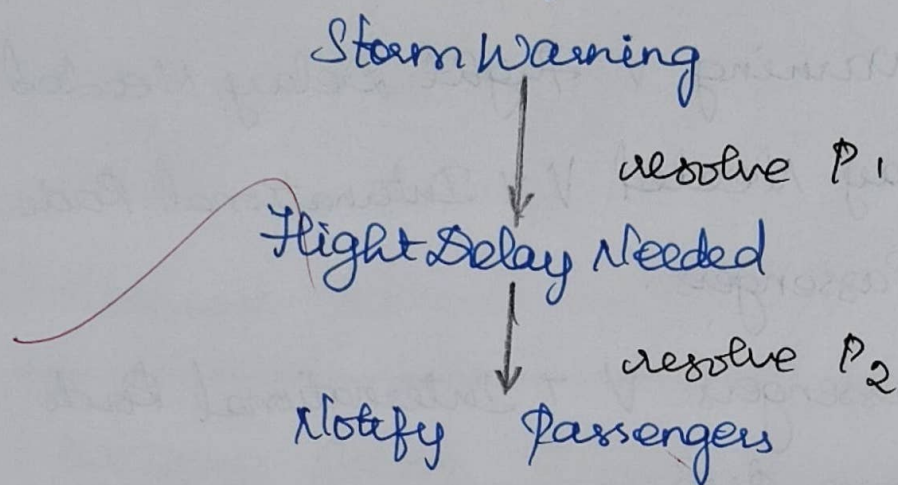
Using Flight Delay Needed:

we obtain: Notify Passengers.

Step 3: Contradiction:

We have: Notify Passengers.

∴ Resolution proof tree:



A.(c) Remove Storm Warning:

Initially, It'll allow Rule 1 to fire.

1. Flight Delay Needed cannot be derived
2. ∴ Notify Passengers cannot be derived
3. Rule 3 also cannot derive

∴ Compliance Risk does not follow from KB.

3:

Notify Passengers 1 International Route $\rightarrow$
Compliance Risk.

Notify Passengers $\vee$ 1 International Route $\vee$
Compliance Risk.

So the CITE KB is

1. Storm Warning $\vee$ Flight Delay Needed.
2. Flight Delay Needed $\vee$ 1 International Route
 $\vee$ Notify Passengers.
3. Notify Passengers $\vee$ 1 International Route
 $\vee$ Compliance Risk.

(b) Resolution proof of Notify Passengers:

Step 1:

from: Storm Warning $\vee$ Flight Delay Needed

and: Storm Warning

resolve to: Flight Delay Needed.

Step 2

P2: Flight Delay Needed $\vee$ 1 International
Route $\vee$ Notify Passengers.

However, the KB does not contain rules

Additional rules needed:

Eg: Critical Condition (x) $\wedge$ Critical Condition (y)
 $\wedge x \neq y \rightarrow \text{Conflict}(x, y)$

and a priority / ordering rule such as:

$\text{Conflict}(x, y) \wedge \text{Higher Priority}(x, y) \rightarrow \text{Route First}(x)$

Thus the current KB is insufficient for simultaneous-patient conflict management.

3. WEATHER BASED FLIGHT SCHEDULING EXPERT SYSTEM:

(a) Conversion to CNF:

Φ_1 : Storm Warning $\rightarrow$ Flight Delay Needed

Using: $P \rightarrow Q \equiv \neg P \vee Q$

We obtain:

$\neg \text{Storm Warning} \vee \text{Flight Delay Needed}$

Φ_2 : Flight Delay Needed $\wedge$ International Route
 $\rightarrow$ Notify Passengers.

$\therefore \neg \text{Flight Delay Needed} \vee \neg \text{International Route} \vee \text{Notify Passengers}.$

Using 3 positive facts, we derive:

Held (Patient B)

Resolution - refutation from

Negate the goal:

$\neg$ Held (Patient B), then resolve with rule 3:

$\neg$ Patient (Patient B) $\vee$ $\neg$ waiting Behind (Patient B,
Patient A) $\vee$ $\neg$ Corridor cleared (Patient A)

$\vee$ Held (Patient B)

Finally: Held (Patient B) and $\neg$ Held (Patient B)

Resolved to: ~~Held~~ (Patient B) is proven.

(c) Two simultaneous critical patients:

The current KB can handle multiple patients only partially.

Eg: If another patient C is also critical:

Patient (Patient C)

Critical Condition (Patient C)

Rule 1 gives:

Priority Corridor (Patient C)

and Rule 2 gives:

Corridor cleared (Patient C)

Facts

Patient (Patient A)

Critical Condition (Patient A)

Patient (Patient B)

Waiting Behind (Patient B, Patient A).

Step 1:

From Rule 1 and Patient (Patient A):

$\neg$ Critical Condition (Patient A) $\vee$ Priority
Corridor (Patient A)

Using:

Critical Condition (Patient A)

We obtain: Priority Corridor (Patient A)

Step 2:

From Rule 2: $\neg$ Priority Corridor (Patient A)

$\vee$ Corridor cleared (Patient A)

Resolving with priority Corridor (Patient A):

Corridor cleared (Patient A)

Step 3:

Use Rule 3 with $x = \text{Patient B}$, $y = \text{Patient A}$,

giving: $\neg$ Patient (Patient B) $\vee \neg$ Waiting Behind

(Patient B, Patient A) $\vee \neg$ Corridor cleared (Patient A)

$\vee$ ~~Not~~ Hold (x).